Evaluation of Similarity-based Next Utterance Prediction Method

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Abstract

Through the use of dialogue history we are able to predict future utterances. A prediction is used in each domain of conversation. In the case of answering questions it is important to predict the answer type from an input. Deciding the answer type is similar to a prediction. In the case of chat, human beings predict the most suitable response based on knowledge and experience acquired. We assume that if an input has a high similarity with an utterance in the dialogue history then a dialogue turn following this utterance becomes the base for predicting the next output. Although current research is largely based upon the prediction of word level, we consider sentence level and utterance level. In this paper, we do prediction experiments using different kinds of corpora. As a result, we confirmed that our similarity is effective in the corpus of a call-center. We describe the experiment and the consideration.

Keywords
Prediction, $TF \cdot AoI$, Spoken dialogue, Similarity

1 Introduction

Recently many dialogue systems are being improved using speech recognition technology. However, it is difficult for such systems to recognize the speech of human beings accurately. Therefore, a number of spoken dialogue systems have been made by combining a speech recognition system with a spoken dialogue system. Such systems have next utterance candidate patterns implemented by hand according to domain limitation. Speech recognition systems usually incorporate a language model and a speech model. The language model generally uses n-gram for calculating the co-occurrence of each word. We believe however that the prediction of utterances as opposed to words is more desirable. Furthermore the prediction of utterances are able to be used in various domains of dialogue. For example in answering questions, a question answering system responds with an answer type selected using utterance prediction. The decision of the answer type is similar to prediction with utterances. This is comparable to human dialogue whereby the most suitable responses during chat are based upon the lifetime knowledge and experience of the participants involved. Therefore our language model will automatically predict and select possible next utterance candidates based upon the similarity between the input and previous utterances contained within the dialogue history. This is similar to predict the most suitable information of partner. Therefore, we try to get next utterance candidates from a dialogue history automatically. Our method predicts the next utterance using a similarity between an input and previous utterance of a dialogue history. Our similarity measure used Amount of Information (AoI) because of easy calculation and word weighting. Our method evaluates the quality of matching words based upon the amount of use. We explain our method and describe a comparison experiment consisting of different corpus and methods.

2 Process

2.1 Input analysis

We explain about a process of our similarity method. Figure 1 shows the process of our method. Since Japanese words are written without a space, we divided into each word by ChaSen[2]. The word frequency of input is calculated.

2.2 Searching history

We call a dialogue history The dialogue history has conversation of time series. Although each utterance in a dialogue ordered by time, each dialogue is randomly assigned in the dialogue history. In the dialogue history, our system

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searches the highest similarity utterance by calculating similarity measure between an input and each utterance. We explain about our method for calculating the similarity in the following paragraph.

2.3 Calculation of similarity

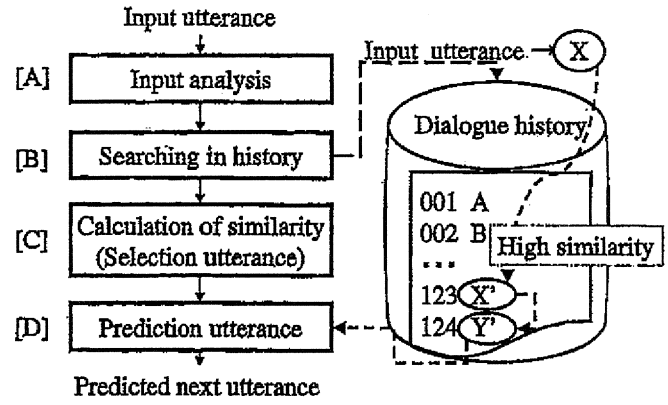


Figure 1: Process of our similarity-based netx utterance prediction.

Our method calculates similarity between two utterances [3]. Here, low frequency words mean important words. AoI is a logarithm of the inverse word frequency; AoI which is short for Amount of Information shows as follows:

$$AoI = -log_2 P(x) = -log_2 \frac{f(x)}{N} \tag{1}$$

N means number of running words, and $f(x)$ means frequency which the word x exists. Each element of the feature is calculated by multiplication between term frequency and amount of information. We call this weight $TF \cdot AoI$. In this paper, we predict next utterance using Euclidean distance based on $TF \cdot AoI$. First, a criteria utterance is divided into each word by the morphological tool. In the case of $D_{input \rightarrow data}$, we call an input utterance the criteria utterance. When "number of difference words" of the criteria utterance is "n", our system calculates by n dimensions. An input ' s $TF(TermFrequency)$ is the word frequency of each word in the input utterance. When different words are $(T_1, T_2, \cdots, T_n)$, features of TF become $TF(TF_1, TF_2, \cdots, TF_n)$. As the features are only the appearance words, TF of the criteria utterance occurs more than once. According to each weight of features, $TF \cdot AoI$ becomes $(TF_1 \cdot AoI(T_1), TF_2 \cdot AoI(T_2), \cdots, TF_n \cdot AoI(T_n))$. The TF of the comparison utterance A becomes $(TF_{a1}, TF_{a2}, \cdots, TF_{an})$. Euclidean Distance between the criteria utterance and the comparison utterance is calculated by $TF \cdot AoI$.

The formulas are as follows:

$$D_{input \rightarrow data} = \sqrt{\sum_{i=1}^{n} (TF_i \cdot AoI(T_i) - TF_{ai} \cdot AoI(T_{ai}))^2} \tag{2}$$

$$D = D_{input \rightarrow data} + \alpha \times D_{data \rightarrow input} \tag{3}$$

$$Similarity = \frac{D_{Standard} - D}{D_{Standard}} \tag{4}$$

When there are no matching words between the criteria utterance and the comparison utterance, the Euclidean Distance becomes $D_{Standard}$. The feature of $D_{input \rightarrow data}$ consists of an input utterance's words and the feature of $D_{data \rightarrow input}$ consists of each utterance's words in the learning corpus. Therefore the highest similarity becomes 1.

2.4 Prediction utterance

We assume that if an input has high similarity with an utterance in the dialogue history, then a dialogue turn following this utterance becomes a base for predicting the next output. Therefore next turn which has high similarity with an utterance comes up for a prediction.

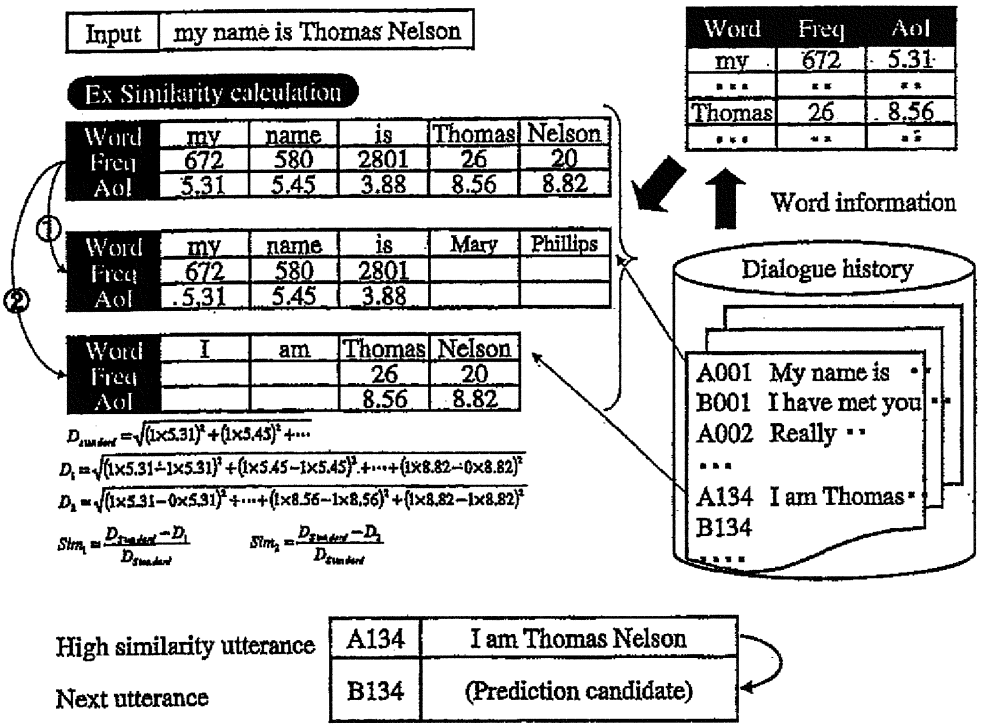


Figure 2: Example of calculation for prediction.

Table 1: Corpora for the prediction experiment.

Domain	Utterance	NRW	NDW	AVG
Chat	844	6,365	1,286	7.54
Call-center	3,376	22,230	1,739	6.58
SLDB	10,512	174,519	4,239	16.6

NRW is the Number of Running Words.
NDW is the Number of Difference Words.
AVG is the average words number per utterance.

3 Experiment

3.1 Experiment corpora

We do prediction experiments using three kinds of corpora. Table.1 shows experiment corpora. We use chat corpus because the corpus resembles spoken dialogue corpus in terms of expressions. This chat corpus is collected from two people using Yahoo messenger[1]. These corpora consist of the chat log of 43 days. The call-center corpus collects the data which wrote down the recorded sound between the agent and a customer. SLDB is short for Speech and Language DataBase[5]. SLDB consists of simulated conversation of a hotel clerk and a visitor. Figure 3 shows examples of these domains. These dialogue examples are divided into each utterance.

3.2 Comparison methods

We compare four methods which consist of $TF \cdot AoI$, String::Similarity, Random and Best. $TF \cdot AoI$ is our method. In addition to a coefficient $\alpha = 0.0$, our method uses coefficient $\alpha = 0.1$ because we confirmed the best coefficient in the preliminary experiment. String::Similarity [4] is distributed in CPAN[2]. This method calculates the similarity measure in consideration of insertion, deletion, and substitution. In this experiment, the Random method and the

[1] http://messenger.yahoo.co.jp/
[2] http://www.cpan.org/

Chat
なにしてるかと思った？！ I was thinking about what you are doing. 自転車とりにいった札幌駅に I went to Sapporo station in order to bring my bicycle. そっか今日は研究しないの？ I see. Don't you research today? してるよ Yes, I do.
Call center
ご伝言お願いしたいんですが。 Can I leave him a message? あっはい。 Yes, of course. あのファックスもらいたいって旨伝えてもらえれば。 I would like him to send a fax. ファックスを。 By fax.
SLDB
はい、分かりました。あと、何名様のご予定ですか。 Very good. And, uh how many people will be in your party? 大人が二名です。 There will be two adults. かしこまりました。どのようなお部屋のタイプをご希望でしょうか。 I see.And, what sort of room would you like? はい、できれば別々がいいので、えーシングルルームでお願いします。 Well, we'd like separate rooms if possible, so singles, please.

Figure 3: Dialogue examples of each domain.

Best method are used as an indicator. The Best method is to select the best F-measure in the learning corpus. The Random method is to select candidates in the learning corpus randomly.

3.3 Evaluation

In this section, we evaluate prediction utterance using F-measure. Recall, Precision and F-measure are follow as.

$$Recall = \frac{Number\ of\ correct\ prediction\ words}{Number\ of\ correct\ words} \tag{5}$$

$$Precision = \frac{Number\ of\ correct\ prediction\ words}{Number\ of\ prediction\ words} \tag{6}$$

$$F-measure = \frac{2 \times Recall \times Precision}{Recall + Precision} \tag{7}$$

Even if human being predicts, it is difficult to predict next utterance perfectly. Therefore we select some prediction candidates. The highest F-measure of the 20 selected utterances is evaluated.

3.4 Results and consideration

Table 2 shows the experiment results which calculate F-measure in each domain. The Best method has known a correct prediction already when other method searches the highest similarity utterance in the dialogue history. Since we use the Best method as an indicator, we do not compare to other methods. In the chat corpus, the highest NCPW rate in four methods was 63.22% by String::Similarity. The highest F-measure in them was 0.2408 by $TF \cdot AoI[\alpha = 0.1]$. In the call-center corpus, the highest NCPW rate in four methods was 70.39% by $TF \cdot AoI[\alpha = 0.1]$. The highest F-measure in them was 0.4697 by $TF \cdot AoI[\alpha = 0.0]$. In the SLDB corpus, the highest NCPW rate in four methods was 73.77% by $TF \cdot AoI[\alpha = 0.1]$. The highest F-measure in them was 0.4529 by $TF \cdot AoI[\alpha = 0.1]$. Figure 4 shows examples of the prediction candidates in the chat experiment. In the experiment, the system found few high similarity utterances in the dialogue history because the chat corpus is very small. However, it is difficult to collect the chat log which consists of two same subjects. Although the F-measure of the Best method is maximum

Table 2: Experiment results.

Corpora	Method	NCPW	NPW	NCW	Precision	Recall	F-measure	$\frac{NCPW}{NCPW_{Best}}$
Chat	Best	2,352	4,794	5,988	0.4906	0.3928	0.4363	100.00%
Chat	$TF \cdot AoI[\alpha = 0.0]$	1,444	6,136	5,988	0.2353	0.2411	0.2382	61.39%
Chat	$TF \cdot AoI[\alpha = 0.1]$	1,440	5,973	5,988	0.2411	0.2405	**0.2408**	61.22%
Chat	String::Similarity	1,487	6,496	5,988	0.2289	0.2483	0.2382	**63.22%**
Chat	Random	1,346	6,252	5,988	0.2153	0.2249	0.2200	57.23%
Call-center	Best	13,200	18,132	21,180	0.7280	0.6232	0.6716	100.00%
Call-center	$TF \cdot AoI[\alpha = 0.0]$	9,224	18,099	21,180	0.5096	0.4355	**0.4697**	69.88%
Call-center	$TF \cdot AoI[\alpha = 0.1]$	9,291	18,723	21,180	0.4962	0.4387	0.4657	**70.39%**
Call-center	String::Similarity	9,265	18,335	21,180	0.5053	0.4374	0.4689	70.19%
Call-center	Random	7,713	16,625	21,180	0.4639	0.3642	0.4080	58.43%
SLDB	Best	15582	22924	26399	0.5902	0.6797	0.6318	100.00%
SLDB	$TF \cdot AoI[\alpha = 0.0]$	11483	24750	26399	0.4350	0.4640	0.4490	73.69%
SLDB	$TF \cdot AoI[\alpha = 0.1]$	11495	24359	26399	0.4354	0.4719	**0.4529**	**73.77%**
SLDB	String::Similarity	11155	24357	26399	0.4226	0.4580	0.4396	71.59%
SLDB	Random	9654	24345	26399	0.3657	0.3965	0.3805	61.96%

NCW is Number of Correct Words. NPW is Number of Prediction Words.
NCPW is Number of Correct Prediction Words.

value, the F-measure was 0.4363. From this result, we confirmed that it was difficult to predict in the chat corpus. Figure 5 shows examples of the prediction candidates in the SLDB experiment. Figure 6 shows examples of the prediction candidates in the call-center experiment. Both the call-center corpus and the SLDB corpus were limited topics. Therefore there is regularity in these conversations. When these similarity methods compare to the Random method, these method improved. However, it is few difference between a similarity method and the Random method in the chat corpus As the call-center result, the NCPW rate of our method was 70.39%(=9,291/13,200), and we confirmed 11.96 points higher increase comparing with the Random method. As the SLDB result, the NCPW rate of our method was 73.77%(=11,495/15,582), and we confirmed 11.81 points higher increase comparing with the Random method. Therefore, we confirmed our similarity method was effective in the restricted dialogue..

Input	受け取った？(Have you received it already?)
Correct prediction	はいってたよ(It was in mailbox.)
TFAOI	
[1]	
Similarity sentence	おきた？ (Have you gotten up?)
Next sentence	そうなのです やっとおきたのです [絵文字] (Yes, I have. I got up at last.)
[2]	
Similarity sentence	ついた？ (Have you received?)
Next sentence	何が (What is it?)

Figure 4: Examples of chat experiment results.

4 Conclusion

We confirmed that it was possible to predict next utterances in the call-center and the SLDB because NCPW (Number of Correct Prediction Words) rate of our similarity calculation method was over 11 points higher than that of the Random method. However, it is hard to predict perfectly because enormous amount of possible expressions in spoken dialogue. It is important to distinguish whether the next utterance should be predicted or not. In addition, the spoken dialogue system sometimes does not have to predict the complete utterance. In other words, we need to use patterns of utterances rather than utterances themselves. Therefore we will try to resolve this problem in the next step.

Input	おそれいります。それでは二十四日は何時ごろご到着の予定でいらっしゃいますか。
(Okay. Thank you very much. And uh what time can we be expecting you on the twenty-fourth?)	
Correct prediction	午後三時ごろになると思います。
(It should be around three in the afternoon.)	

String::Similarity	
Similarity sentence	それから十三日の何時ごろ取りにいらっしゃいますか。
(So, what time on the thirteenth would you like to come and pick it up?)	
Next sentence	そちらの営業時間はどうなっているんですか。
(well I'd like to know about your office hours.)	

TFAOI	
Similarity sentence	はい、恐れ入ります、あの時間は何時ごろいらっしゃるご予定でございますか。
(Could I ask, about uh what time what part of the day would you like to go there?)	
Next sentence	ランチに行きますので、一時から三時でお願いします。
(oh We'd like to go for lunch.|So that would be from around one o'clock till three o'clock.)	

Figure 5: Examples of SLDB experiment results.

Input	[redacted]と言います。(My name is [redacted].)
Correct prediction	[redacted]様で。(Mr. [redacted])

String::Similarity	
[1]	
Similarity sentence	[redacted]と申します。(I am T[redacted])
Next sentence	[redacted]様で。(Mr. K[redacted].)
[2]	
Similarity sentence	:いいです。 (That's ok.)
Next sentence	:あよろしいですか。(I see.)

TFAOI	
[1]	
Similarity sentence	[redacted]カマーケティングの[redacted]と申します。 (I am S[redacted] in K[redacted] marketing.)
Next sentence	はい、いつもお世話になっております。 (Hello, Thank you for calling.)
[2]	
Similarity sentence	[redacted]様、大変お待たせしております。 (How are you, Mr. [redacted]?)
Next sentence	はい。 (Fine thank you.)

Figure 6: Examples of call-center experiment results.

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